

# IoT Asset Tracking and Sensor Monitoring System

## Database Design Document

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## Project Overview

The intention of this project is to design and implement a relational db and web-based application for an IoT Asset Tracking and Sensor Monitoring company. The system will provide a centralised platform for managing the IoT devices, firmware versions, device types, and groups used to structure deployed devices. The system will focus on managing the metadata associated with the IoT assets, while the web-based application will allow users to manage firmware versions and groups but still ensuring data integrity by its normalized MS SQL Server db.

## Problem Statement

The company manufactures and deploys multiple IoT devices used for tracking and environmental monitoring. The objective of this project is therefore to design a scalable database that is capable of managing these relationships, while reducing data duplication and maintaining integrity.

## Functional Requirements

- Store IoT Devices
- Store Firmware Versions
- Associate Devices with Firmware
- Organize Devices into Groups
- Allow Groups to contain other Groups
- Store multiple Device Types
- Allow multiple Devices to share the same Firmware Version
- Allow multiple Devices to share the same Device Type
- Provide functionality for: Firmware & Groups

## Non-Functional Requirements

- Maintainable
- Scalable
- Reliable
- Efficient
- Usable
- Future Proof

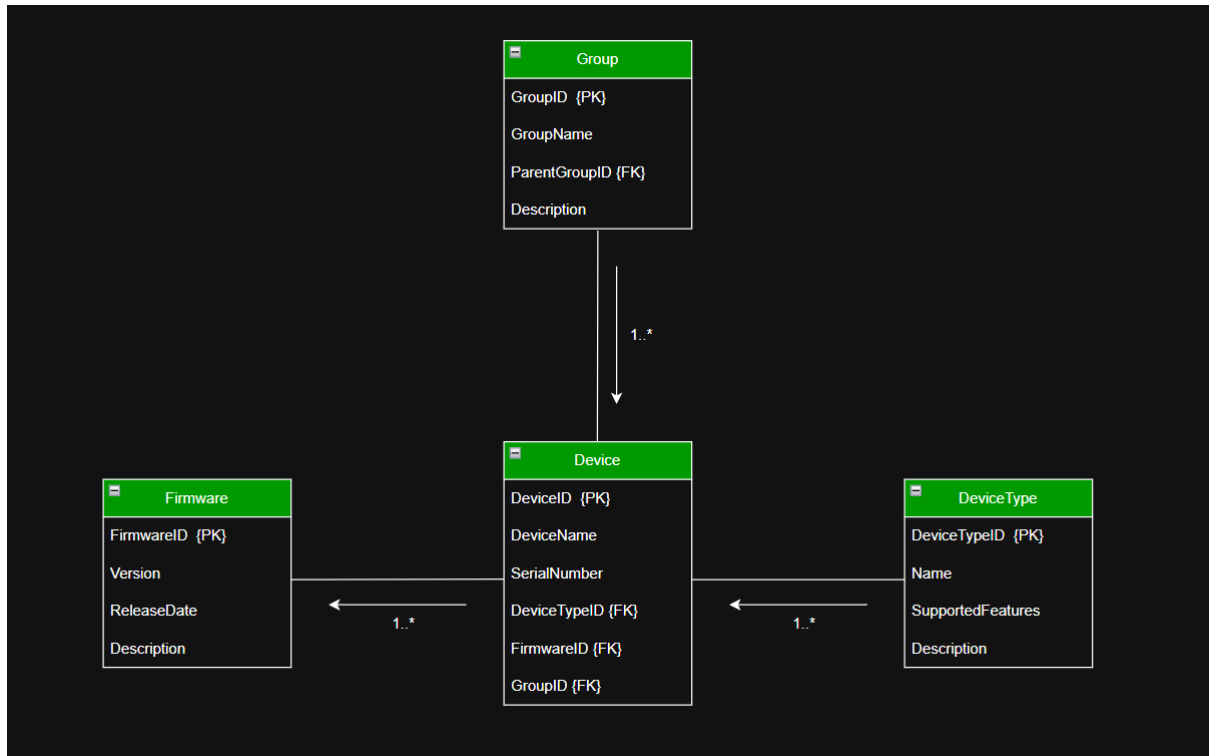
## Assumptions

- Each Device belongs to one Device Type.
- Each Device runs one Firmware Version.
- Multiple Devices can share the same Firmware Version.
- Multiple Devices can share the same Device Type.
- A Device belongs to only one Group.
- A Group may contain zero or many child Groups.

- Groups may exist without Devices.
- Firmware Versions are reusable.

### Entity Identification and ERD

| Entity     | Purpose                 |
|------------|-------------------------|
| Device     | Physical IoT Device     |
| Firmware   | Store firmware versions |
| DeviceType | Device category         |
| Group      | Organises devices       |



### Relationships

- Firmware – Device: One-to-Many (many devices can use the same firmware version)
- DeviceType – Device: One-to-Many (many physical devices can be manufactured from the same product model)
- Group – Device: One-to-Many (one deployment location can contain multiple devices)
- Group – Group: One-to-Many (for structuring)

## DB Design

### (Device)

| Column       | Type     | Description   |
|--------------|----------|---------------|
| DeviceID     | INT      | PK            |
| DeviceName   | NVARCHAR | Name          |
| SerialNumber | NVARCHAR | Unique Number |
| DeviceTypeID | INT      | FK            |
| FirmwareID   | INT      | FK            |
| GroupID      | INT      | FK            |

### (DeviceType)

| Column            | Type     | Description                 |
|-------------------|----------|-----------------------------|
| DeviceTypeID      | INT      | PK                          |
| Name              | NVARCHAR | Product Type                |
| Description       | NVARCHAR | Product Description         |
| SupportedFeatures | NVARCHAR | Summary of all capabilities |

### (Firmware)

| Column      | Type     | Description        |
|-------------|----------|--------------------|
| FirmwareID  | INT      | PK                 |
| Version     | NVARCHAR | Firmware Version   |
| Description | NVARCHAR | Release Date       |
| ReleaseDate | Date     | Full Release Notes |

### (Group)

| Column        | Type     | Description       |
|---------------|----------|-------------------|
| GroupID       | INT      | PK                |
| GroupName     | NVARCHAR | Group Name        |
| Description   | NVARCHAR | Group Description |
| ParentGroupID | INT      | Parent Group      |

## Conclusion

This proposed database design ensures a scalable, normalized, and maintainable foundation/platform for managing all IoT devices, firmware versions, and deployed groups, while also reducing data redundancy and keeping flexible enough to allow for future enhancement without the need of significant changes.